

MPEC Approach

C.Conlon - thanks to late great Che-Lin Su

Grad IO

Often faced with extremum estimator problems in econometrics (ML, GMM, MD, etc.) that look like:

$$\hat{\theta} = \arg \max_{\theta} Q_n(\theta), \quad \theta \in \Theta$$

Many economic problems contain constraints, such as: market clearing (supply equals demand), consumer's consume their entire budget set, or firm's first order conditions are satisfied. A natural way to represent these problems is as constrained optimization.

MPEC

$$\hat{\theta} = \arg \max_{\theta, P} Q_n(\theta, P), \quad \text{s.t.} \quad \Psi(P, \theta) = 0, \quad \theta \in \Theta$$

Fixed Point / Implicit Solution

In much of the literature the tradition has been to express the solutions $\Psi(P, \theta) = 0$ implicitly as $P(\theta)$:

$$\hat{\theta} = \arg \max_{\theta} Q_n(\theta, P(\theta)), \quad \theta \in \Theta$$

MPEC

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In much of the literature the tradition has been to express the solutions $\Psi(P, \theta) = 0$ implicitly as $P(\theta)$:

$$\hat{\theta} = \arg \max_{\theta} Q_n(\theta, P(\theta)), \quad \theta \in \Theta$$

Probably you were taught some things that weren't quite right

- ▶ Fewer parameters → easier problems to solve!
- ▶ Reformulate problems with fixed points or implicit solutions to **concentrate out** parameters.
- ▶ But sometimes this makes the problem more complicated (saddle points, complicated Hessians, etc.)

MPEC says do the opposite:

- ▶ Add lots of parameters
- ▶ But add them with simple constraints (linear or quadratic).
- ▶ Idea: Make the Hessian as close to constant, block diagonal, sparse, etc. as possible.

Mostly this is in response to change in technology: nonlinear solvers supporting large sparse Hessians.

- ▶ Bus repairman sees mileage x_t at time t since last overhaul
- ▶ Repairman chooses between overhaul and normal maintenance

$$u(x_t, d_t, \theta^c, RC) = \begin{cases} -c(x_t, \theta^c) & \text{if } d_t = 0 \\ -(RC + c(0, \theta^c)) & \text{if } d_t = 1 \end{cases}$$

- ▶ Repairman solves DP:

$$V_{\theta}(x_t) = \sum_{f_t, f_{t+1}, \dots} E \left\{ \sum_{j=t}^{\infty} \beta^{j-t} [u(x_j, f_j, \theta) + \varepsilon_j(f_j)] | x_t \right\}$$

- ▶ Structural parameters to be estimated $\theta = (\theta^c, RC, \theta^p)$.
- ▶ Coefficients of cost function $c(x, \theta^c) = \theta_1^c x + \theta_2^c x^2$
- ▶ Transition probabilities in mileages $p(x_{t+1} | x_t, d_t, \theta^p)$

- Data: time series $(x_t, d_t)_{t=1}^T$
- Likelihood function

$$\mathcal{L}(\theta) = \prod_{t=2}^T P(d_t|x_t, \theta^c, RC) p(x_t|x_{t-1}, d_{t-1}, \theta^p)$$

$$\text{with } P(d|x, \theta^c, RC) = \frac{\exp[u(x, d, \theta^c, RC) + \beta EV_\theta(x, d)]}{\sum_{d' \in \{0,1\}} \exp[u(x, d', \theta^c, RC) + \beta EV_\theta(x', d)]}$$

$$EV_\theta(x, d) = T_\theta(EV_\theta)(x, d)$$

$$\equiv \int_{x'=0}^{\infty} \log \left[\sum_{d' \in \{0,1\}} \exp[u(x, d', \theta^c, RC) + \beta EV_\theta(x', d)] \right] p(dx'|x, d, \theta^p)$$

- Outer Loop: Solve Likelihood

$$\max_{\theta \geq 0} \mathcal{L}(\theta) = \prod_{t=2}^T \mathbb{P}(d_t | x_t, \theta^c, RC) p(x_t | x_{t-1}, d_{t-1}, \theta^p)$$

- Convergence test: $\|\nabla_{\theta} \mathcal{L}(\theta)\| \leq \epsilon_{out}$
- Inner Loop: Compute expected value function EV_{θ} for a given θ
- EV_{θ} is the implicit expected value function defined by the Bellman equation or the fixed point function

$$EV_{\theta} = T_{\theta}(EV_{\theta})$$

- Convergence test: $\|EV_{\theta}^{(k+1)} - EV_{\theta}^{(k)}\| \leq \epsilon_{in}$
- Start with contraction iterations and polish with Newton Steps

- ▶ Inner-loop error propagates into outer-loop function and derivatives
- ▶ NFXP needs to solve inner-loop exactly each stage of parameter search
 - to accurately compute the search direction for the outer loop
 - to accurately evaluate derivatives for the outer loop
 - for outer loop to converge!
- ▶ Stopping rules: choosing inner-loop and outer-loop tolerance
 - inner loop can be slow: contraction mapping is linearly convergent
 - tempting to loosen inner loop tolerance ϵ_{in} (such as $1e-6$ or larger!).
 - Outer loop may not converge with loose inner loop tolerance.
 - check solver output message
 - tempting to loosen outer loop tolerance ϵ_{out} to promote convergence ($1e-3$ or larger!).

- ▶ $\mathcal{L}(EV(\theta, \epsilon_{in}), \theta)$ the programmed outer loop objective function
- ▶ L : the Lipschitz constant (like modulus) of inner-loop contraction mapping
- ▶ Analytic derivatives $\nabla_{\theta} \mathcal{L}(EV(\theta, \epsilon_{in}), \theta)$ is provided: $\epsilon_{out} = O(\frac{L}{1-L} \epsilon_{in})$
- ▶ Finite-difference derivatives are used: $\epsilon_{out} = O(\sqrt{\frac{L}{1-L}} \epsilon_{in})$

- Form the augmented likelihood function for data $X = (x_t, d_t)_{t=1}^T$

$$\mathcal{L}(EV, \theta; X) = \prod_{t=2}^T P(d_t | x_t, \theta^c, RC) p(x_t | x_{t-1}, d_{t-1}, \theta^p)$$

$$\text{with } P(d|x, \theta^c, RC) = \frac{\exp[u(x, d, \theta^c, RC) + \beta EV(x, d)]}{\sum_{d' \in \{0,1\}} \exp[u(x, d', \theta^c, RC) + \beta EV(x', d)]}$$

- Rationality and Bellman equation imposes a relationship between θ and EV

$$EV = T(EV, \theta)$$

- Solve constrained optimization problem

$$\max_{(\theta, EV)} \mathcal{L}(EV, \theta; X)$$

$$\text{subject to } EV = T(EV, \theta)$$

The previous approach solves the problem “on the grid”.

- ▶ x_t takes on discrete values.
- ▶ We only evaluate $EV(x_t, i_t)$ at values on the grid.
- ▶ We don't evaluate $EV(x_t, i_t)$ and values between $[x_s, x_{s+1}]$.

If we did we would have to **interpolate**.

- ▶ Macroeconomists tend to use **cubic splines**
- ▶ Could use global orthogonal polynomials $EV(x) \approx a_0 + a_1 b_1(x) + a_2 b_2(x) + \dots$
 - Advantage is that solving polynomial equation for $EV = T(EV, \theta)$ is pretty easy.
 - Can easily switch to continuous state space without any additional complications: $f(x_{t+1}|x_t)$ must also be continuous.

Results

β	Imple.	Parameters						MSE
		RC	θ_{11}	θ_{31}	θ_{32}	θ_{33}	θ_{34}	
	true	11.726	2.457	0.0937	0.4475	0.4459	0.0127	
0.975	MPEC1	12.212	2.607	0.0943	0.4473	0.4454	0.0127	3.111
		(1.613)	(0.500)	(0.0036)	(0.0057)	(0.0060)	(0.0015)	–
	MPEC2	12.212	2.607	0.0943	0.4473	0.4454	0.0127	3.111
		(1.613)	(0.500)	(0.0036)	(0.0057)	(0.0060)	(0.0015)	–
	NFXP	12.213	2.606	0.0943	0.4473	0.4445	0.0127	3.123
		(1.617)	(0.500)	(0.0036)	(0.0057)	(0.0060)	(0.0015)	–
0.980	MPEC1	12.134	2.578	0.0943	0.4473	0.4455	0.0127	2.857
		(1.570)	(0.458)	(0.0037)	(0.0057)	(0.0060)	(0.0015)	–
	MPEC2	12.134	2.578	0.0943	0.4473	0.4455	0.0127	2.857
		(1.570)	(0.458)	(0.0037)	(0.0057)	(0.0060)	(0.0015)	–
	NFXP	12.139	2.579	0.0943	0.4473	0.4455	0.0127	2.866
		(1.571)	(0.459)	(0.0037)	(0.0057)	(0.0060)	(0.0015)	–

Results

β	Imple.	Parameters						MSE
		RC	θ_{11}	θ_{31}	θ_{32}	θ_{33}	θ_{34}	
	true	11.726	2.457	0.0937	0.4475	0.4459	0.0127	
0.985	MPEC1	12.013	2.541	0.0943	0.4473	0.4455	0.0127	2.140
		(1.371)	(0.413)	(0.0037)	(0.0057)	(0.0060)	(0.0015)	–
	MPEC2	12.013	2.541	0.0943	0.4473	0.4455	0.0127	2.140
		(1.371)	(0.413)	(0.0037)	(0.0057)	(0.0060)	(0.0015)	–
	NFXP	12.021	2.544	0.0943	0.4473	0.4455	0.0127	2.136
		(1.368)	(0.411)	(0.0037)	(0.0057)	(0.0060)	(0.0015)	–
0.990	MPEC1	11.830	2.486	0.0943	0.4473	0.4455	0.0127	1.880
		(1.305)	(0.407)	(0.0036)	(0.0057)	(0.0060)	(0.0015)	–
	MPEC2	11.830	2.486	0.0943	0.4473	0.4455	0.0127	1.880
		(1.305)	(0.407)	(0.0036)	(0.0057)	(0.0060)	(0.0015)	–
	NFXP	11.830	2.486	0.0943	0.4473	0.4455	0.0127	1.880
		(1.305)	(0.407)	(0.0036)	(0.0057)	(0.0060)	(0.0015)	–

Results

β	Imple.	Parameters						MSE
		RC	θ_{11}	θ_{31}	θ_{32}	θ_{33}	θ_{34}	
	true	11.726	2.457	0.0937	0.4475	0.4459	0.0127	
0.995	MPEC1	11.819	2.492	0.0942	0.4473	0.4455	0.0127	1.892
		(1.308)	(0.414)	(0.0036)	(0.0057)	(0.0060)	(0.0015)	–
	MPEC2	11.819	2.492	0.0942	0.4473	0.4455	0.0127	1.892
		(1.308)	(0.414)	(0.0036)	(0.0057)	(0.0060)	(0.0015)	–
	NFXP	11.819	2.492	0.0942	0.4473	0.4455	0.0127	1.892
		(1.308)	(0.414)	(0.0036)	(0.0057)	(0.0060)	(0.0015)	–

Results

β	Imple.	Runs Conv.	CPU Time (in sec.)	# of Major Iter.	# of Func. Eval.	# of Contrac. Mapping Iter.
0.975	MPEC1	1240	0.13	12.8	17.6	—
	MPEC2	1247	7.9	53.0	62.0	—
	NFXP	998	24.6	55.9	189.4	$1.348e + 5$
0.980	MPEC1	1236	0.15	14.5	21.8	—
	MPEC2	1241	8.1	57.4	70.6	—
	NFXP	1000	27.9	55.0	183.8	$1.625e + 5$
0.985	MPEC1	1235	0.13	13.2	19.7	—
	MPEC2	1250	7.5	55.0	62.3	—
	NFXP	952	42.2	61.7	227.3	$2.658e + 5$
0.990	MPEC1	1161	0.19	18.3	42.2	—
	MPEC2	1248	7.5	56.5	65.8	—
	NFXP	935	70.1	66.9	253.8	$4.524e + 5$
0.995	MPEC1	965	0.14	13.4	21.3	—
	MPEC2	1246	7.9	59.6	70.7	—
	NFXP	950	111.6	58.8	214.7	$7.485e + 5$

Results

```
KNITRO 5.2.0: alg=1
opttol=1.0e-6
feastol=1.0e-6
```

Problem Characteristics

Objective goal: Maximize

Number of variables: 207

 bounded below: 6

 bounded above: 201

 bounded below and above: 0

 fixed: 0

 free: 0

Number of constraints: 202

 linear equalities: 1

 nonlinear equalities: 201

 linear inequalities: 0

 nonlinear inequalities: 0

 range: 0

Number of nonzeros in Jacobian: 2785

Number of nonzeros in Hessian: 1620

Results

Final Statistics

Final objective value	=	-2.35221126396447e+03
Final feasibility error (abs / rel)	=	1.33e-15 / 1.33e-15
Final optimality error (abs / rel)	=	1.00e-08 / 6.71e-10
# of iterations	=	12
# of CG iterations	=	0
# of function evaluations	=	13
# of gradient evaluations	=	13
# of Hessian evaluations	=	12
Total program time (secs)	=	0.10326 (0.097 CPU time)
Time spent in evaluations (secs)	=	0.05323

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KNITRO 5.2.0: Locally optimal solution.
objective -2352.211264; feasibility error 1.33e-15
12 major iterations; 13 function evaluations

BLP 1995

The estimator solves the following mathematical program:

$$\min_{\theta_2} g(\xi(\theta_2))' W g(\xi(\theta_2)) \quad \text{s.t.}$$

$$g(\xi(\theta_2)) = \frac{1}{N} \sum_{\forall j,t} \xi_{jt}(\theta_2)' z_{jt}$$

$$\xi_{jt}(\theta_2) = \delta_j(\theta_2) - x_{jt}\beta - \alpha p_{jt}$$

$$s_{jt}(\delta(\theta_2), \theta_2) = \int \frac{\exp[\delta_j(\theta_2) + \mu_{ij}]}{1 + \sum_k \exp[\delta_j(\theta_2) + \mu_{ik}]} f(\mu | \theta_2)$$

$$\log(S_{jt}) = \log(s_{jt}(\delta(\theta_2), \theta_2)) \quad \forall j, t$$

The estimation algorithm is generally as follows:

1. Guess a value of nonlinear parameters θ_2
2. Compute $s_{jt}(\delta, \theta_2)$ via integration
3. Iterate on $\delta_{jt}^{h+1} = \delta_{jt}^h + \log(S_{jt}) - \log(s_{jt}(\delta^h, \theta_2))$ to find the δ that satisfies the share equation
4. IV Regression δ on observable X and instruments Z to get residual ξ .
5. Use ξ to construct $g(\xi(\theta_2))$.
6. Possibly construct other errors/instruments from supply side.
7. Construct GMM Objective

The idea is that $\delta(\theta_2)$ is an implicit function of the nonlinear parameters θ_2 . And for each guess we find that implicit solution for reduce the parameter space of the problem. But the Jacobian:

$$\frac{\partial \xi_t}{\partial \theta_2}(\theta_2) = - \left[\frac{\partial \mathbf{s}_t}{\partial \delta_t}(\theta_2) \right]^{-1} \left[\frac{\partial \mathbf{s}_t}{\partial \theta_2}(\theta_2) \right] \text{ is complicated to compute.}$$

BLP-MPEC

The estimator solves the following mathematical program:

$$\min_{\Sigma, \alpha, \beta, \xi} g(\xi)' W g(\xi) \quad \text{s.t.}$$

$$g(\xi) = \frac{1}{N} \sum_{\forall j, t} \xi'_{jt} z_{jt}$$

$$s_{jt}(\Sigma, \alpha, \beta, \xi) = \sum_i w_i \frac{\exp[x_{jt}\beta + \xi_{jt} - \alpha p_{jt} + (\Sigma \cdot \nu_{il}) x_{jt}]}{1 + \sum_k \exp[x_{kt}\beta + \xi_{kt} - \alpha p_{kt} + (\Sigma \cdot \nu_{il}) x_{kt}]}$$

$$\log(S_{jt}) = \log s_{jt}(\Sigma, \alpha, \beta, \xi) \quad \forall j, t$$

- ▶ Expand the parameter space of the nonlinear search to include α, β, ξ
- ▶ Don't have to solve for ξ except at the end.
- ▶ No implicit functions of θ_2 and $\left[\frac{\partial \mathbf{s}_t}{\partial \sigma}(\sigma, \alpha, \beta, \xi) \right]$ is straightforward (no matrix inverse!).

Nevo Results

	Nevo	BLP-MPEC	EL
Price	-28.189	-62.726	-61.433
σ_p	0.330	0.558	0.524
σ_{const}	2.453	3.313	3.143
σ_{sugar}	0.016	-0.006	0
σ_{mushy}	0.244	0.093	0.085
$\pi_{p,inc}$	15.894	588.206	564.262
$\pi_{p,inc2}$	-1.200	-30.185	-28.930
$\pi_{p,kid}$	2.634	11.058	11.700
$\pi_{c,inc}$	5.482	2.29084	2.246
$\pi_{c,age}$	0.2037	1.284	1.37873
GMM	29.3611	4.564	
EL			-17422
Time	28 s	12s	19s

Another Example: Empirical Likelihood

Consider the GMM moment condition $\mathbb{E}[g(Z_i, \theta)] = 0$

- ▶ A common example is $\mathbb{E}[z_i'(y_i - x_i\beta)] = 0$
- ▶ Now consider re-weighting the data with ρ_i so that $\sum_i \rho_i g(Z_i, \theta) = 0$
 - We also need that $\sum_i \rho_i = 1$ and $\rho_i \in (0, 1)$ (ie: ρ is a valid probability measure)
 - Idea: penalize distance between ρ_i and $\frac{1}{n}$ with $f(\rho)$
- ▶ Idea: your moments always hold (because economic theory)
- ▶ How much would you have to re-sample data?

Another Example: (Generalized) Empirical Likelihood

$$\begin{aligned} \min_{\theta, \rho} \sum_i f(\rho_i) \quad \text{s.t.} \quad & \sum_i \rho_i \cdot g(Z_i, \theta) = 0 \\ & \sum_i \rho_i = 1 \quad \rho_i \geq 0 \end{aligned}$$

Choices of $f(\rho_i)$:

- ▶ Empirical Likelihood: $f(\rho_i) = -\log \rho_i$ [This is NPMLE]
- ▶ Continuously Updating GMM (CUE): $f(\rho_i) = \rho_i^2$ [Derivation is **not** obvious]
- ▶ Exponential Tilt: $f(\rho_i) = \rho_i \log \rho_i$ [has some “robustness” properties]
- ▶ Adversarial Method of Moments: Conlon, Kim, Manresa, Li (2025)
- ▶ See Kitamura’s Handbook Chapter for more details (and convex analysis) or Newey Smith (2004) for asymptotic bias.

Empirical Likelihood: NFXP Solution

Start with the Lagrangian:

$$\mathcal{L} = \sum_{i=1}^n \log \rho_i + \lambda \left(1 - \sum_{i=1}^n \rho_i \right) - n \gamma' \sum_{i=1}^n \rho_i \cdot g(Z_i, \theta)$$

Take derivatives with respect to ρ and θ and Lagrange multipliers to get γ and concentrate out $\rho_i(\theta)$

$$\hat{\gamma}(\theta) = \arg \min_{\gamma \in \mathbb{R}^q} - \sum_{i=1}^n \log (1 + \gamma' g(Z_i, \theta))$$

$$\hat{\rho}_i(\theta) = \frac{1}{n (1 + \hat{\gamma}(\theta)' g(Z_i, \theta))}, \quad \hat{\lambda} = n$$

The resulting dual is a saddle-point problem:

$$\hat{\theta}_{EL} = \arg \max_{\theta \in \Theta} \ell_{NP}(\theta) = \arg \max_{\theta \in \Theta} \min_{\gamma \in \mathbb{R}^q} - \sum_{i=1}^n \log (1 + \gamma' g(Z_i, \theta))$$

But the primal (MPEC) problem is much easier... unless $g(Z_i, \theta)$ is trivial.

Thanks!
